

● EASY

● GEOMETRY AND TRIGONOMETRY

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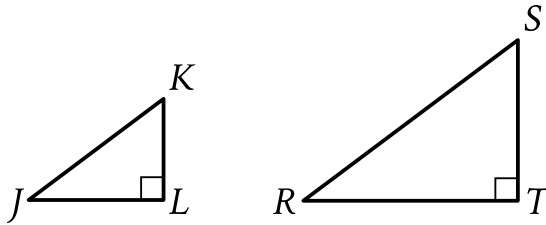
# Area and volume

# 03

9 questions · ~14 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Right triangle upper  $J$  upper  $K$  upper  $L$  is labeled as follows:
  - Angle upper  $L$  is a right angle.
- Right triangle upper  $R$  upper  $S$  upper  $T$  is labeled as follows:
  - Angle upper  $T$  is a right angle.
- A note indicates the figures are not drawn to scale.

In the figure shown, triangle  $JKL$  is similar to triangle  $RST$ , where  $J$  corresponds to  $R$  and  $K$  corresponds to  $S$ . The length of  $\bar{JK}$  is 15, and the perimeter of triangle  $JKL$  is 36. The length of  $\bar{RS}$  is 135. What is the perimeter of triangle  $RST$ ?

- A. 324
- B. 540
- C. 2,916
- D. 8,100

Triangle R has an area of 80 square centimeters  $\text{cm}^2$ . Square S has side lengths of 4 cm. What is the total area of triangle R and square S, in  $\text{cm}^2$ ?

- A. 42
- B. 44
- C. 84
- D. 96

A rectangle has a length of 13 and a width of 6. What is the perimeter of the rectangle?

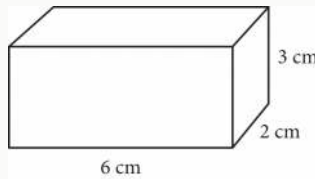
- A. 12
- B. 26
- C. 38
- D. 52

A rectangle has an area of 63 square meters and a length of 9 meters. What is the width, in meters, of the rectangle?

- A. 7
- B. 54
- C. 81
- D. 567

What is the area of a rectangle with a length of 17 centimeters cm and a width of 7 cm?

- A.  $24 \text{ cm}^2$
- B.  $48 \text{ cm}^2$
- C.  $119 \text{ cm}^2$
- D.  $576 \text{ cm}^2$



The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume  $V$  of a right rectangular prism is  $\ell wh$ , where  $\ell$  is the length of the prism,  $w$  is the width of the prism, and  $h$  is the height of the prism. What is the volume, in cubic centimeters, of the prism?

- A. 36
- B. 24
- C. 12
- D. 11

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

A triangle has a base length of 40 centimeters and a height of 90 centimeters. What is the area, in square centimeters, of the triangle?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



The area of a rectangle is 630 square inches. The length of the rectangle is 70 inches. What is the width, in inches, of this rectangle?

- A. 9
- B. 70
- C. 315
- D. 560

The table gives the perimeters of similar triangles  $TUV$  and  $XYZ$ , where  $TU$  corresponds to  $XY$ . The length of  $TU$  is 6.

|                | Perimeter |
|----------------|-----------|
| Triangle $TUV$ | 50        |
| Triangle $XYZ$ | 150       |

What is the length of  $XY$ ?

- A. 2
- B. 6
- C. 18
- D. 56